Transient Two-Dimensional Heat Conduction in a Rectangular Bar: Explicit FTCS and ADI Solutions

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The transient temperature field in a long copper bar with fixed wall temperatures is computed with the explicit forward-time, centered-space (FTCS) scheme and the alternating direction implicit (ADI) scheme. The steady state is compared with the analytical methods on a grid of 31 x 41. With a time step of 0.2s both schemes reach steady-state, which is defined as the total variation in temperature per step being less than $0.01\,^{\circ}\mathrm{C}$ defined as CONSS after about 410 s of simulated time. For both methods the max error is $0.2874\,^{\circ}\mathrm{C}$ near the hot corners where the wall discontinuity exists, root-mean-square error of $0.0260\,^{\circ}\mathrm{C}$, mean error of $-0.0011\,^{\circ}\mathrm{C}$. With a timestep of 1.0 s, FTCS diverges, as its stability limit predicts, ADI converges in 496 steps. The steady-state wall heat flux satisfies the overall energy balance of the system within $0.1\,\%$.

Nomenclature

L, W	=	bar length in x and width in y, m
T	=	temperature, °C
T_0	=	initial interior temperature, °C
T_1, T_2, T_3, T_4	=	bottom, left, top, and right wall temperatures, °C
α	=	thermal diffusivity, $\mathrm{m^2\,s^{-1}}$
k	=	thermal conductivity, $\mathrm{W\,m^{-1}\,{}^{\circ}C^{-1}}$
t	=	time, s
Δt	=	time step, s
Δx, Δy	=	grid spacing in x and y, m
i, j	=	grid indices in x and y
n	=	time level
d_x, d_y	=	FTCS diffusion numbers, $\alpha\Delta t/\Delta x^2$ and $\alpha\Delta t/\Delta y^2$
d_1, d_2	=	ADI half-step coefficients, $\alpha\Delta t/(2\Delta x^2)$ and $\alpha\Delta t/(2\Delta y^2)$
G	=	amplification factor
TV	=	total variation, sum of $\lvert T_{i,j}^{n+1} - T_{i,j}^{n}\rvert$ over interior nodes, °C
CONSS	=	steady-state tolerance on TV, °C
m	=	series term index
q_{in}	=	heat flux into the bar, $\mathrm{W\,m^{-2}}$

I. Introduction

This report presents solutions to Problems 3.5 through 3.8 of Hoffmann and Chiang [1]. Problem 3.5 describes transient heat conduction in a long rectangular copper bar, initially at a uniform temperature, the four sides are held at constant temperatures. The transient solution is computed with the explicit forward-time, centered-space (FTCS) scheme, the alternating direction, implicit (ADI) scheme, or the fractional step method (FSM), and the steady state is compared with the analytical series solution. The steady-state heat flux is evaluated along the sides $y = 0$, $y = W$, and $x = 0$. The ADI method was selected over FSM. Finally, the accuracy, stability, and efficiency of the three schemes are compared.

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II. Problem Statement

The geometry shown in Fig. 1 defines the axis system as x-right and y-up with the length and width of the bar defined as L and W and respectively. The figure defines the indices i and j in base one whereas in the solution provided all indices are given in base 0 notation with the ends of the bar defined as IM and JM respectively. Fig. 1

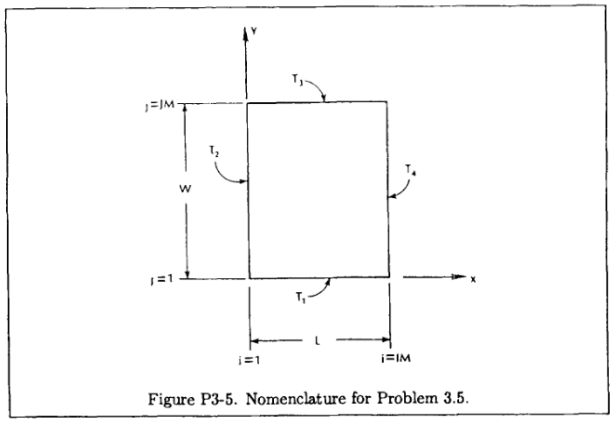


Figure P3-5. Nomenclature for Problem 3.5.

Fig. 1 Geometry, axes, and boundary conditions of the bar (after Ref. [1]).

The temperature in the bar satisfies the two-dimensional heat conduction equation

$$\frac{\partial T}{\partial t} = \alpha \left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} \right) \tag{1}$$

on $0 \le x \le L$, $0 \le y \le W$, with $T = T_0$ initially and fixed wall temperatures $T(x, 0) = T_1$, $T(0, y) = T_2$, $T(x, W) = T_3$, and $T(L, y) = T_4$. It is assumed that the solution has reached steady state if the total variation in temperature from one time point to the next is less than CONSS. CONSS is given as 0.01 °C in the problem statement.

III. Numerical Methods

A. Explicit FTCS Scheme

$$T_{i,j}^{n+1} = T_{i,j}^{n} + d_x \left(T_{i+1,j}^{n} - 2T_{i,j}^{n} + T_{i-1,j}^{n} \right) + d_y \left(T_{i,j+1}^{n} - 2T_{i,j}^{n} + T_{i,j-1}^{n} \right) \tag{2}$$

The scheme is stable for $d_x + d_y \le 1/2$.

B. Alternating Direction Implicit Scheme

$$-d_1 T_{i-1,j}^{*} + (1 + 2d_1) T_{i,j}^{*} - d_1 T_{i+1,j}^{*} = d_2 T_{i,j-1}^{n} + (1 - 2d_2) T_{i,j}^{n} + d_2 T_{i,j+1}^{n} \tag{3}$$

$$-d_2 T_{i,j-1}^{n+1} + (1 + 2d_2) T_{i,j}^{n+1} - d_2 T_{i,j+1}^{n+1} = d_1 T_{i-1,j}^{*} + (1 - 2d_1) T_{i,j}^{*} + d_1 T_{i+1,j}^{*} \tag{4}$$

C. Thomas Algorithm

The ADI method as discussed splits the pentadiagonal system into a pair of tridiagonal systems of equations. The naive approach to solving a system of equations would be to employ gaussian elimination which has a big O complexity of $O(n^3)$. To efficiently solve the special case of a tridiagonal system Eq. (5) the Thomas Algorithm [Thomas 1949] was developed. Hoffmann and Chiang [1] demonstrates the same algorithm in Appendix B. under a different name. This approach takes advantage of the zero symmetry about the diagonal to solve the system in $O(n)$ complexity. The notation below matches the code provided.

$$a_k x_{k-1} + b_k x_k + c_k x_{k+1} = d_k, \qquad k = 0, 1, \ldots, n-1, \qquad a_0 = c_{n-1} = 0 \tag{5}$$

$$\begin{bmatrix} b_0 & c_0 & & & \\ a_1 & b_1 & c_1 & & \\ & \ddots & \ddots & \ddots & \\ & & a_{n-2} & b_{n-2} & c_{n-2} \\ & & & a_{n-1} & b_{n-1} \end{bmatrix} \begin{bmatrix} x_0 \\ x_1 \\ \vdots \\ x_{n-2} \\ x_{n-1} \end{bmatrix} = \begin{bmatrix} d_0 \\ d_1 \\ \vdots \\ d_{n-2} \\ d_{n-1} \end{bmatrix} \tag{6}$$

Where the a_k,b_k,c_k is known and d_k is unknown. For each time step the inner nodes of a row form a system of equations Eq. (6). The algorithm begins with a linear forwards sweep handling the first equation which does not have a_k by using equations Eq. (7). Then the rest of the rows forward sweep is solved of the form Eq. (8).

$$s_0 = \frac{c_0}{b_0}, \qquad x'_0 = \frac{d_0}{b_0} \tag{7}$$

$$\left.\begin{aligned} \delta_k &= b_k - a_k\, s_{k-1} \\ s_k &= \frac{c_k}{\delta_k}, \qquad k \le n-2 \\ x'_k &= \frac{d_k - a_k\, x'_{k-1}}{\delta_k} \end{aligned}\right\} \qquad k = 1, 2, \ldots, n-1 \tag{8}$$

The loop completes over the inner nodes and then back substitutes following Eq. (9).

$$x_{n-1} = x'_{n-1}, \qquad x_k = x'_k - s_k\, x_{k+1}, \qquad k = n-2, n-3, \ldots, 0 \tag{9}$$

The algorithm returns with the solved rows unknown d_k.

D. Analytical Steady-State Solution

The analytical steady state solution provided by the problem statement can be simplified for the case of $T_2 = T_4 = 0$ as Eq. (10). The solution below is continuous and was evaluated at the same grid of nodes as the solvers for consistency of comparison. Numerical stability of the hyperbolic sin had to be taken into account due to double precision floating point overflow. The identity $\sinh a/\sinh b = e^{a-b}(1 - e^{-2a})/(1 - e^{-2b})$, for $0 \le a \le b$, enables the stable calculation of Eq. (10).

$$T(x, y) = \sum_{m \text{ odd}} \frac{4}{m\pi} \sin\!\left(\frac{m\pi x}{L}\right) \frac{T_1 \sinh[m\pi(W - y)/L] + T_3 \sinh(m\pi y/L)}{\sinh(m\pi W/L)} \tag{10}$$

E. Convergence and Wall Heat Flux

$$q_{\text{in}} = -k\, \nabla T \cdot \hat{n}_{\text{in}}, \qquad \left.\frac{\partial T}{\partial y}\right|_{y=0} \approx \frac{-3T_{i,0} + 4T_{i,1} - T_{i,2}}{2\Delta y} \tag{11}$$

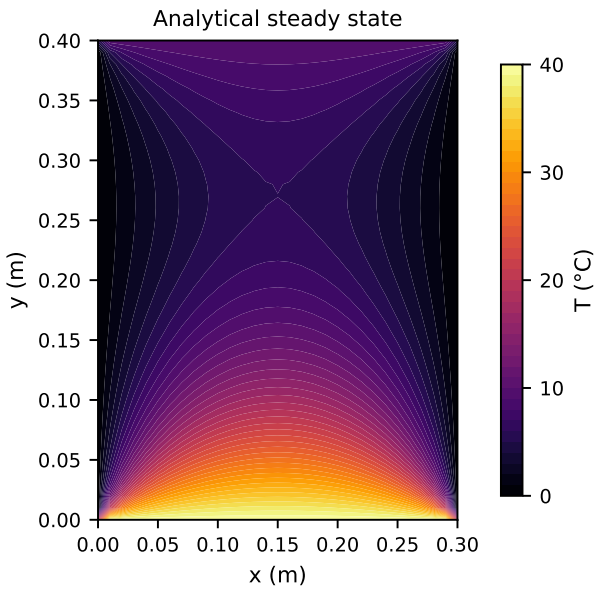


Fig. 2 Analytical steady-state temperature field.

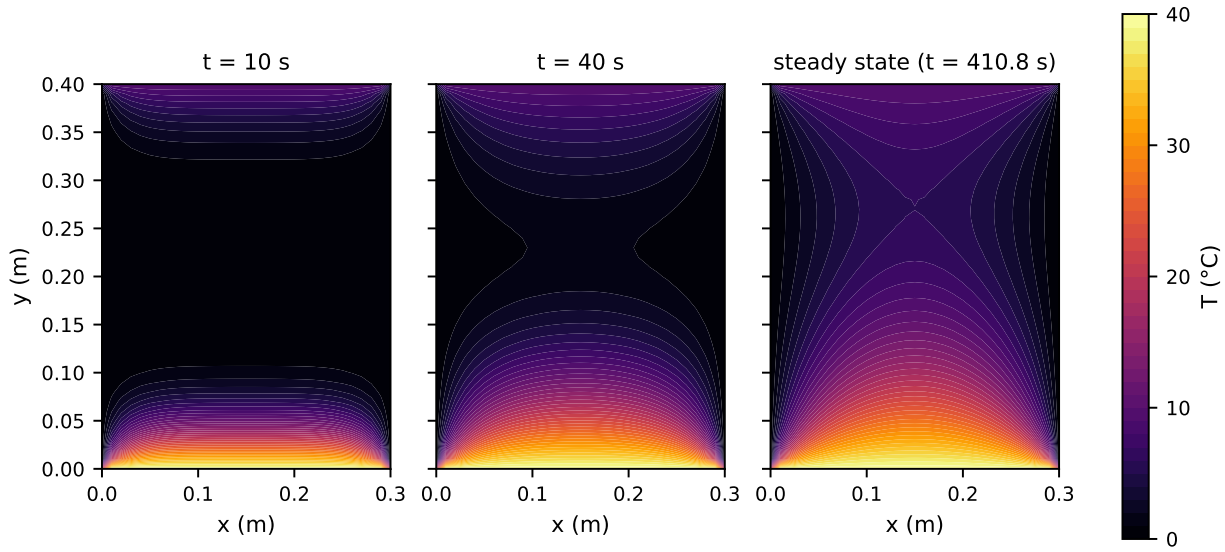


Fig. 3 FTCS temperature field at $t = 10\,\mathrm{s}$, $t = 40\,\mathrm{s}$, and steady state, $\Delta t = 0.2\,\mathrm{s}$.

IV. Results

A. Analytical Baseline

B. FTCS

1. Transient Solution

2. Stability

3. Temperature Histories and Convergence

4. Steady State Compared with the Analytical Solution

C. ADI

1. Transient Solution

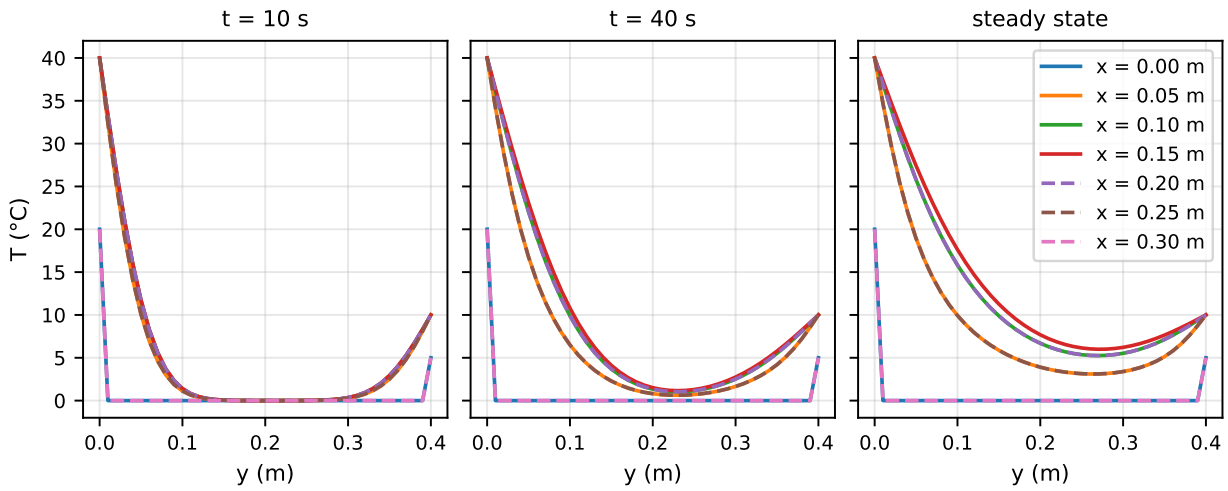


Fig. 4 FTCS temperature versus y at $x = 0$ to $0.30\,\mathrm{m}$ in steps of $0.05\,\mathrm{m}$, $\Delta t = 0.2\,\mathrm{s}$.

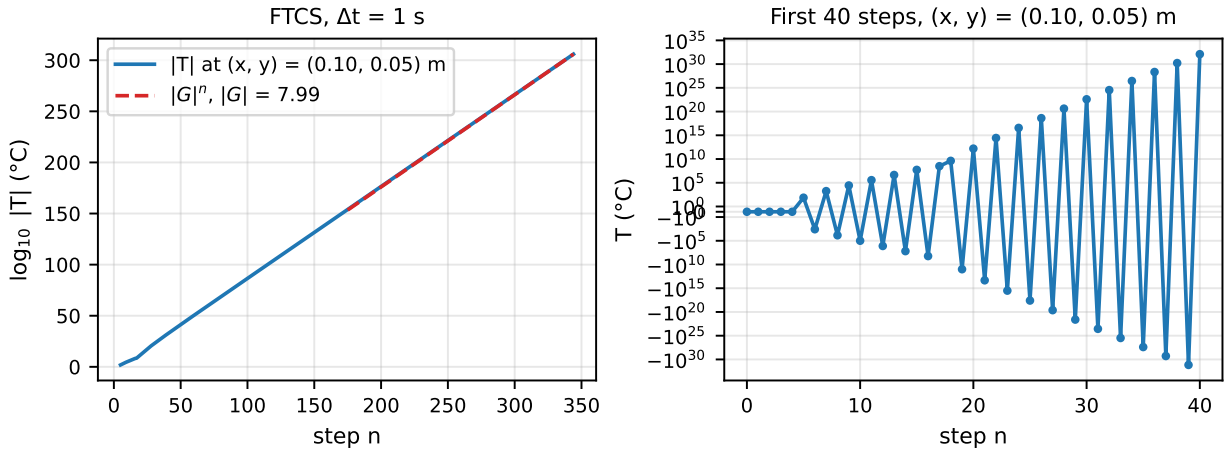


Fig. 5 FTCS with $\Delta t = 1.0\,\mathrm{s}$: growth of the probe temperature compared with the amplification factor $|G|^n$, and the first 40 steps.

and supported the debugging of the author's implementations of the FTCS, ADI, Thomas, and analytical solutions. It also supported the initial LaTeX structure, validation of LaTeX equations, figure captions of this report and educated on the AIAA format, which the author revised. The author verified all results, wrote the remaining text, and is responsible for all of the content provided.

References

[1] Hoffmann, K. A., and Chiang, S. T., *Computational Fluid Dynamics*, 4th ed., Vol. 1, Engineering Education System, Wichita, KS, 2000.

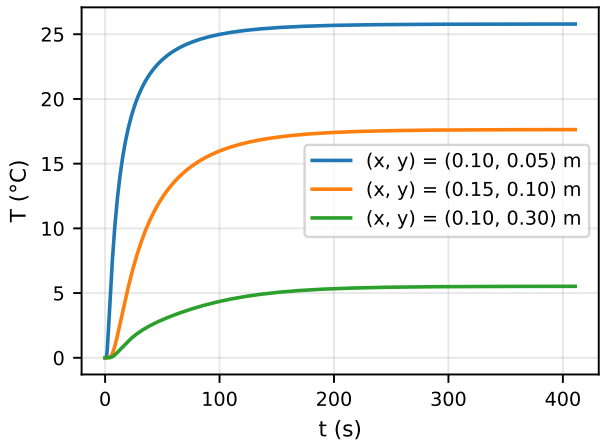


Fig. 6 FTCS temperature histories at the three probe locations, $\Delta t = 0.2\,\mathrm{s}$.

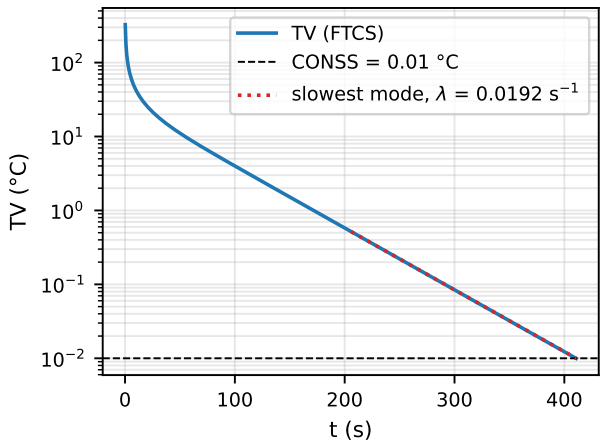


Fig. 7 FTCS total variation versus time with the steady-state tolerance and the slowest-mode decay rate.

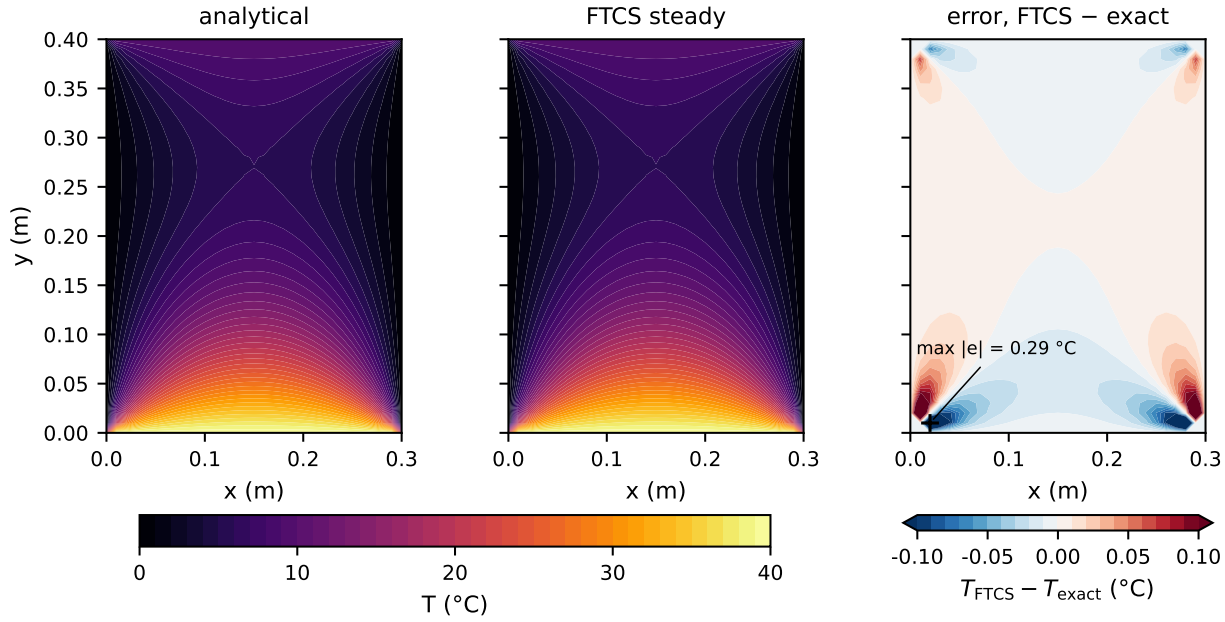


Fig. 8 Analytical and FTCS steady-state fields and their difference.

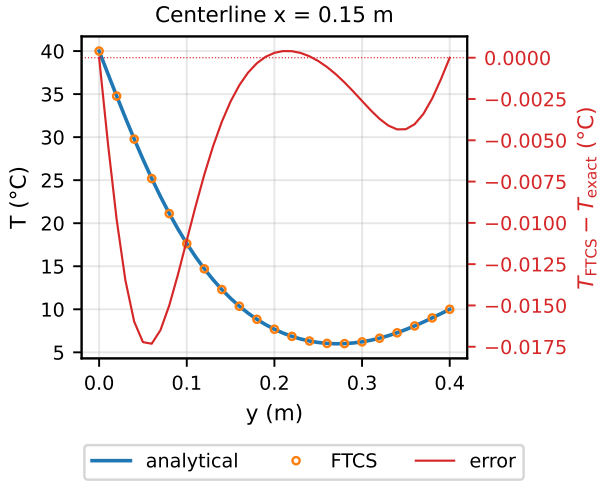


Fig. 9 Steady-state temperature along the centerline $x = 0.15$ m, with the FTCS error.

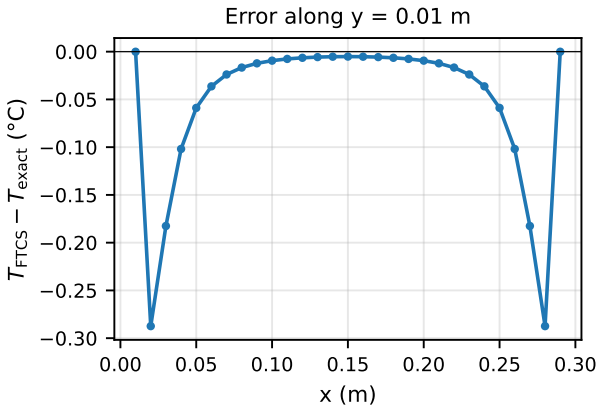


Fig. 10 FTCS steady-state error along the first interior row, $y = 0.01$ m.

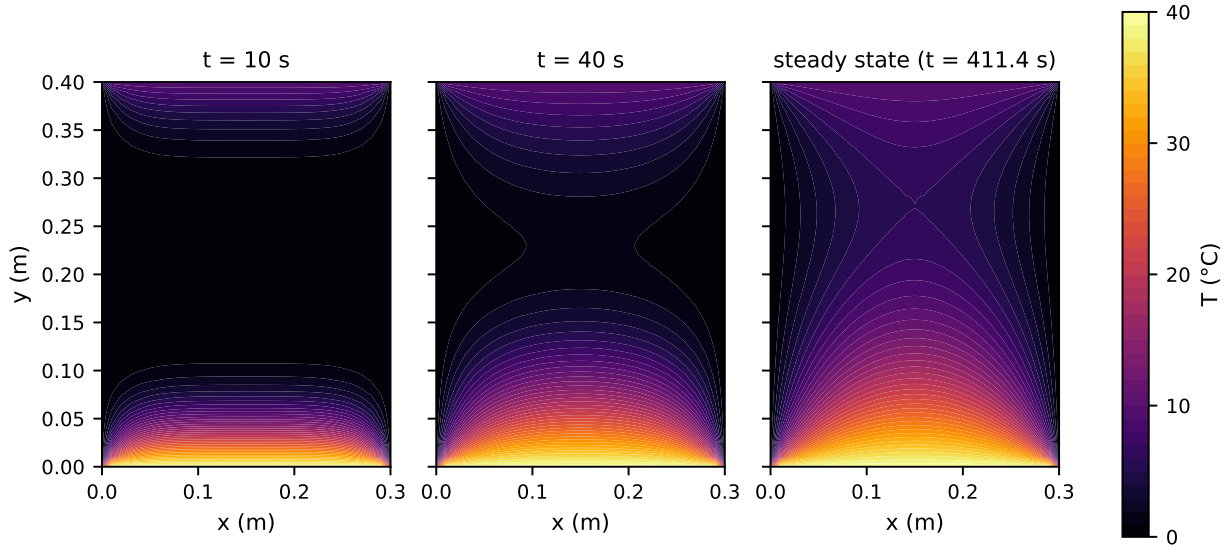


Fig. 11 ADI temperature field at $t = 10$ s, $t = 40$ s, and steady state, $\Delta t = 0.2$ s.

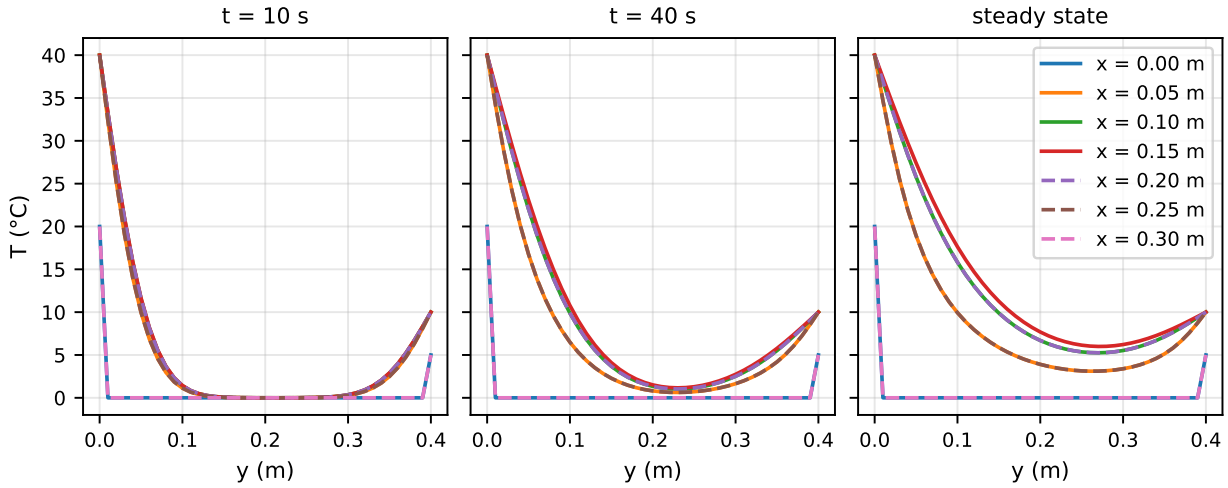


Fig. 12 ADI temperature versus y at $x = 0$ to $0.30\,\mathrm{m}$ in steps of $0.05\,\mathrm{m}$, $\Delta t = 0.2\,\mathrm{s}$.

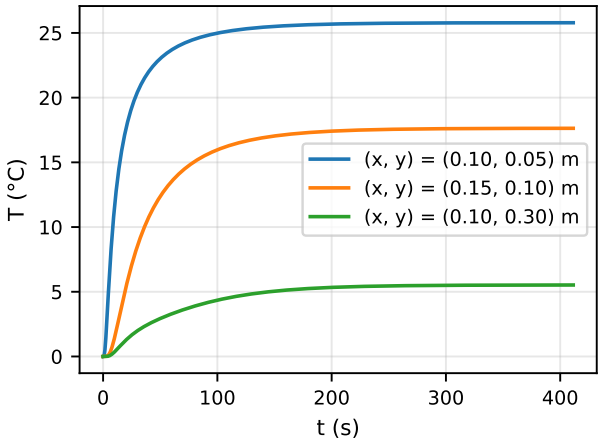


Fig. 13 ADI temperature histories at the three probe locations, $\Delta t = 0.2\,\mathrm{s}$.

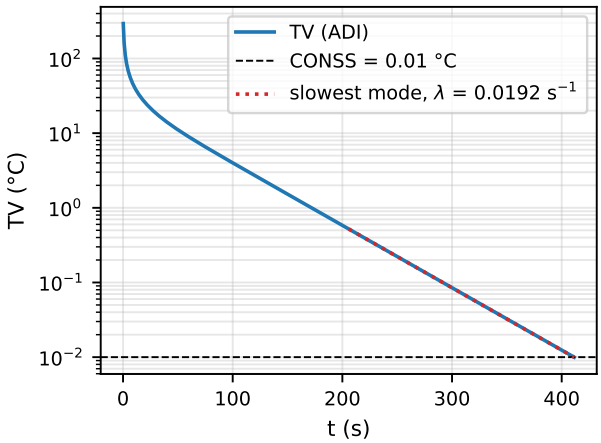


Fig. 14 ADI total variation versus time with the steady-state tolerance and the slowest-mode decay rate.

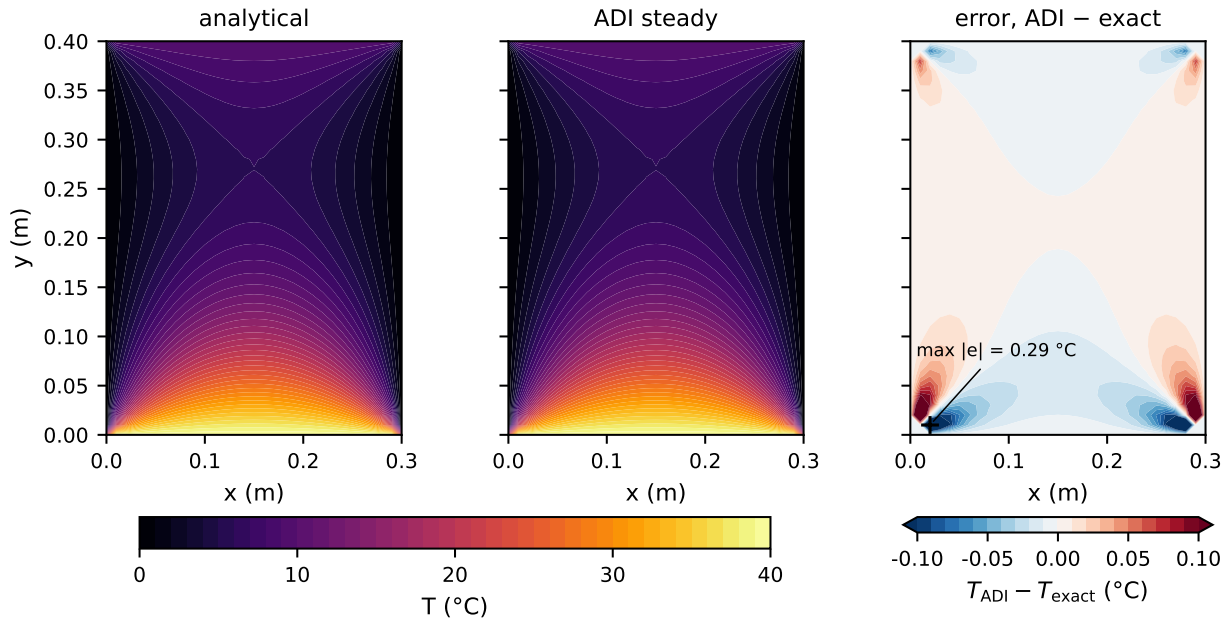


Fig. 15 Analytical and ADI steady-state fields and their difference.

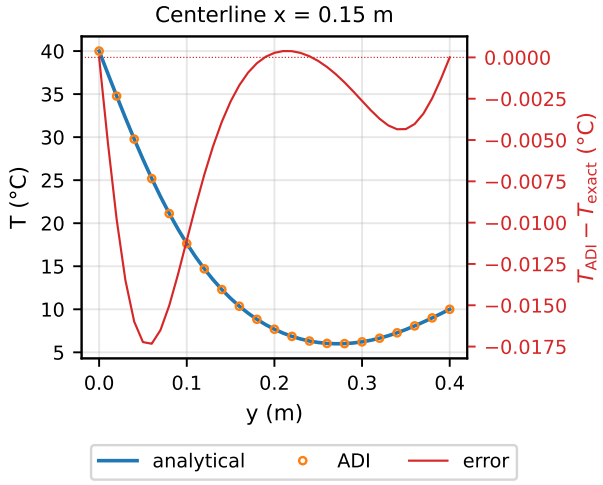


Fig. 16 Steady-state temperature along the centerline $x = 0.15\,\mathrm{m}$, with the ADI error.

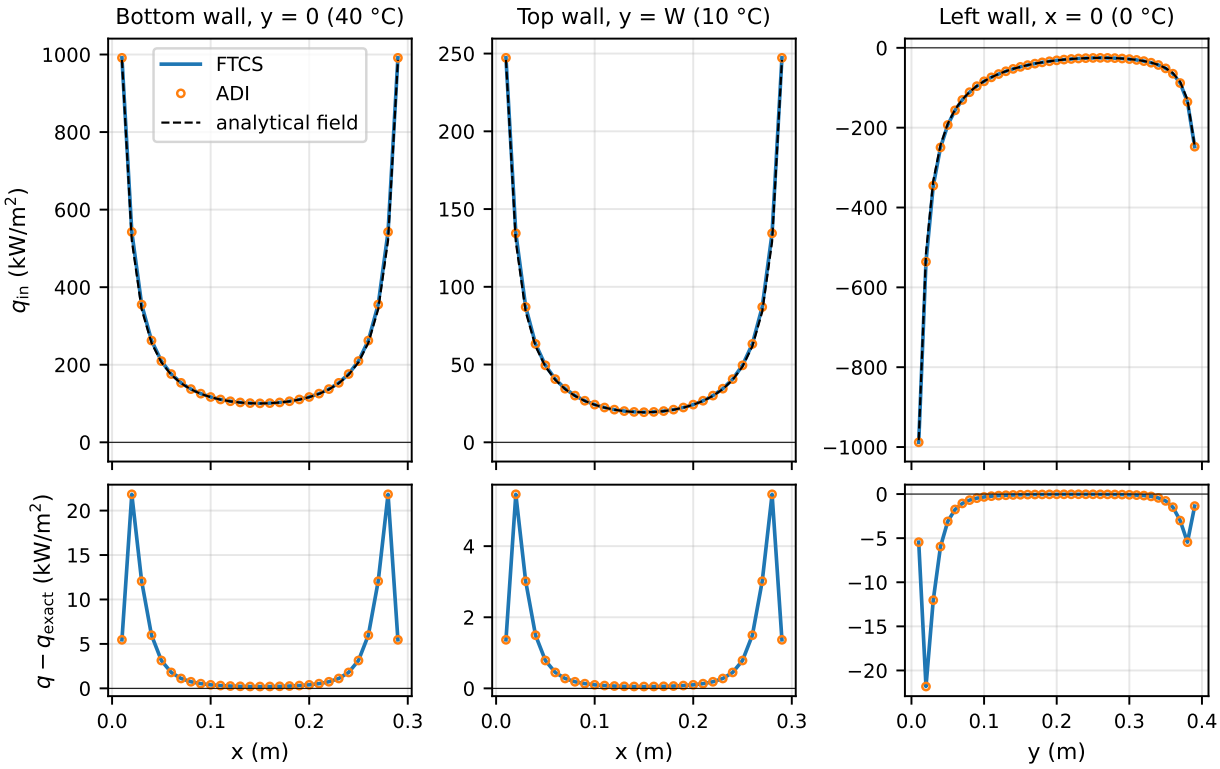


Fig. 17 Steady-state heat flux into the bar along the bottom, top, and left walls (top row) and the difference from the analytical field (bottom row). Corners are excluded; the flux there is unbounded.